"Value proposition canvas - to develop a digital solution to alleviate loneliness among the elderly in Sweden”

Background problem statement: In What Way Might we reduce the feeling of loneliness in elderly people who have difficulties taking part in activities outside their home due to reduced mobility.

Product: Robot dog - and what value and social situations it can be a digital solution for the user.

Annex 1: Customer Profile Description Template (Blue addition is AI feedback)

JOBS (rank jobs: important – insignificant)	What is your persona trying to achieve? rank: important – insignificant Fundamental objectives: To stop the progression of feelings of loneliness. To avoid progression to social isolation. To increase involvement and engagement in one’s social life Instrumental objectives: 1.Staying connected to family and friends 2.Be involved in the lives of family and friends - Keep track of upcoming and previous events and conversations. 3. To pass time and chat with. 4. Maintaining daily routines without feeling managed or bossed around. 5. Preserving cognitive health and memory
Assumptions	What are our assumptions behind jobs? 1.Few people admit to being lonely 2. People can feel connected and care for a robot 3. People want to feel connected to friends, family and society 4. People want to be able to use platforms and digital tools that are used by others in society 5. People want to feel needed and wanted in society 6. Assumption: Elderly people value small moments of social interaction throughout the day 7. Assumption: A social companion can facilitate human relationships rather than replace them 8. Assumption: Elderly people want to contribute to relationships, not only receive attention

PAINS (rank pains: extreme - moderate)	What are your persona’s pains? – rank: extreme – moderate 1.Limited experience of digital solutions 2.High threshold for testing new digital solutions 3.Fear of making mistakes, “pushing the wrong button”. 4. Fear of being scammed 5. Dexterity problems 6. Reduced hearing and vision 7. Reduced mobility. 8. No regular activities and reduced social connections after leaving the work force. 9. Difficult to keep up with rapid digitalization and evolution of e.g. AI. 10. Limited ability to use new ways of communicating and connecting with other people. 11. Shrinking circle of friends to socialize with due to illness and old age. 12. Limited financial means as retirees. 13. Fear of patronization or being treated like a child. 14. Anxiety over changing technology (e.g., software updates altering how the dog behaves). 15. Fear of tripping over physical objects in the home.

Assumptions	What are our assumptions behind pains? 1. Rapid global digital evolution 2. More individualistic living society 3. Natural aging causes (gradual) decrease in functional status 4. There is a willingness to pay a total of 5 000 SEK for a robot dog among family members of the elderly. 5. The main barrier is not only technology, but the opportunity to connect

GAINS (rank gains: essential – nice to have)	What are your persona's gains? – rank: essential – nice to have 1. Social Connections/Feeling connected 2. Overall - life quality 3. Increased self confidence and security 4. Independency from surrounding 5. Less stress about keeping track of daily activities. 6. Reassurance for family members without feeling like a burden. 7. A sense of purpose (e.g., "taking care" of the dog gives a reason to structure the day)
Assumptions	What are our assumptions behind gains? 1. Communication would be facilitated if one could talk directly to a robot dog with access to all contacts (instead of communicating on a smart phone). This would be a preferred option for our target group. 2. Easy communication would increase independency. 3. Easy communication would increase social connections and reduce feeling of loneliness.

Annex 2: Value Map Description Template

Product/ Service	What is the core of our PRODUCT / SERVICE? And, how does it connect to the JOBS your customer is trying to do? We are offering a robot dog for people with limited mobility who are struggling to maintain their connectivity to society in general and family and friends in particular. Our dog is always available for a chat or a cuddle, but it is also poking and nudging the customer to encourage real life relations. It will want to go outside and may be set to seek out other robot dog owners. It will keep track of activities going on in the local area and suggest a visit, based on the customers revealed preferences. The dog offers a modern solution to vocal communication, replacing the smartphone. It will suggest calling different contacts, e.g. on their birthday. Low-maintenance charging dock (magnetic or self-docking) so users don't struggle with cords. Tactile touch zones (petting response) for grounded, screen-free interaction.
Assumptions	What are the assumptions behind this design? 1. The dog's physical presence will reduce sentiments of loneliness 2. The customer will have more social contacts if this is facilitated.

Pain Relievers	Are you eliminating pains? And, how is your product/service doing this? (PAIN RELIEVERS) 1. Limited digital experience, high threshold and fear of getting it wrong: The dog is self explanatory. There are no buttons to push. 2. Fear of being scammed: The dog has limited functions and can't make financial transactions. It can't send its own messages. 3. There are no buttons to push, the dog is voice controlled, which relieves the problems with dexterity and poor vision. 4. As the customer ages, the dog will draw on earlier preferences and experiences to encourage the user to maintain important social contacts. 5. Transparent privacy protection (data stays local; non-intrusive listening). 6. Adaptive speech recognition tuned for aging voices, tremors, or quiet speaking.
Assumptions	What are the assumptions behind this design? 1. The dog will have a health benefit, making it eligible for public funding 2. Assumption: Knowing who is available lowers the barrier to initiating contact 3. Assumption: The solution can create new social connections, not only maintain existing ones 4. Assumption: Voice interaction makes spontaneous social contact easier

Gain Creators	Are you addressing gains? And, how is your product/service doing this? (GAIN CREATORS) 1. Social connections - both new and old - are facilitated. The dog will encourage the user to call and/or visit old friends. It may also suggest meeting up with other robot dog owners. This may require some safety features such as registered users. 2. Quality of life will increase. 3. Self-confidence and security. Our dog adapts to the customer's needs and its personality evolves over time. It uses evidence based methods to build the customer's sense of self-worth and -confidence. It can also call for help if the user experiences acute medical problems. 4. Reciprocal care dynamic: The dog provides a gentle routine (e.g., "Good morning, time for coffee") that gives structure to the day while making the user feel needed.
Assumptions	What are the assumptions behind this design?

	1. Quality of life is associated with social contacts and physical activity. 2. Increased self-confidence will encourage the user to take part in social activities. 3. The dog will act as a personal assistant, keeping track of appointments and activities. This will increase independence and relieve stress. 4. Communication is facilitated when the robot dog can be instructed to call someone and there is no need to use a phone to find contact details and to make the call.

Annex 3: Test Card Description

Video: How to design/use test cards? https://www.youtube.com/watch?v=cW46ySJmLD8

Test Card 1 Customer Profile	Assumption: (mention assumption here) - We believe that: There is a willingness to pay a total of 5 000 SEK for a robot dog among family members of the elderly. - To verify that, we will market our dog on Facebook with a visible price - And measure how many sign up to buy the prototype at the price of 5 000 SEK. - We are right if 100 people sign up.
Test Card 2 Customer Profile	Assumption: (mention assumption here) - Communication would be facilitated if one could talk directly to a robot dog with access to all contacts (instead of communicating on a smart phone). This would be a preferred option for our target group. - To verify that, we will interview ten elderly people about how they would feel about talking through a robot dog. - And measure how likely people say they would be to use this function - unlikely, likely, very likely. - We are right if 7 people say they are likely or very likely to use this function.
Test Card 3 Customer Profile	Assumption: The main barrier is not only technology, but the opportunity to connect - We believe that: Elderly people with limited mobility experience loneliness not only because communication technology can be difficult to use, but because they do not always know who is available to talk, do not want to disturb family members, and have too few natural opportunities for spontaneous social interaction. - To verify that, we will: Interview at least 10 elderly people with limited mobility about situations when they wanted company or someone to talk to but did not contact anyone. - And measure: How many mention barriers such as “I didn't want to bother them,” “they were probably busy,” “I didn't know who to call,” or “there was nobody available.” - We are right if: At least 50% of the interviewees identify one or more of these barriers.
Test Card 4 Customer Profile	Assumption: Elderly people value small moments of social interaction throughout the day

	- We believe that: Several small social interactions during the day such as receiving a voice message, exchanging a photograph, having a five-minute conversation, or receiving a spontaneous greeting can create a stronger feeling of social connection than relying only on occasional longer phone calls or visits. - To verify that, we will: Interview elderly people about what types of social interaction they value and present examples of small “social moments” that could occur throughout their day. - And measure: How many say that they would appreciate several small interactions throughout the day and which types of interaction they would value most. - We are right if: At least 70% say that they would appreciate one or more forms of small, spontaneous social interaction during their day.
Test Card 1 Value Map	Assumption: (mention assumption here) - Communication is facilitated when the robot dog can be instructed to call someone and there is no need to use a phone to find contact details and to make the call. - To verify that, we will interview twenty people with reduced dexterity and/or vision - And measure how many report at least 5 on a 1-7 Likert scale that this would facilitate their communication. - We are right if at least 75% give a positive answer. -
Test Card 2 Value Map	Assumption: (mention assumption here) - We believe that the dog will have a health benefit, making it eligible for public funding - To verify that, we will send a survey to 50 municipalites/cities in Sweden, expecting 40 to reply - And measure how many report being likely or certain to pay up to 5 000 SEK for a robot dog - We are right if at least 20 cities or 50% give a positive reply.
Test Card 3 Value Map	Assumption: Knowing who is available lowers the barrier to initiating contact - We believe that: Elderly people would be more likely to initiate social contact if they knew that the other person was available and wanted to talk. - To verify that, we will: Present a simple “Who's free?” concept in which family members, friends or trusted contacts can indicate when they are available for a conversation. The elderly person could simply ask the robot dog, “Who's available?” without using a smartphone.

	- And measure: How many elderly interviewees say that knowing someone is available would make them more comfortable initiating a conversation. - We are right if: At least 70% say they would be more likely to contact someone if they knew beforehand that the person was available.
Test Card 4 Value Map	Assumption: A social companion can facilitate human relationships rather than replace them - We believe that: Elderly people would value a simple social companion, such as a robot dog, whose primary purpose is to help create contact with other people rather than replacing human companionship with conversations with an AI. - To verify that, we will: Present two alternatives to elderly interviewees: A. A robot dog mainly designed to talk with the elderly person itself. B. A robot dog mainly designed to help the elderly person connect with family, friends, neighbors and other people. - And measure: Which concept they prefer and why. - We are right if: At least 60% prefer the robot as a facilitator of human contact rather than primarily as an artificial conversation partner.
Test Card 5 Value Map	Assumption: Voice interaction makes spontaneous social contact easier - We believe that: Elderly people with reduced mobility, dexterity, vision or confidence using digital technology would find it easier to initiate social interaction through simple spoken commands rather than navigating a smartphone or tablet. - To verify that, we will: Demonstrate example interactions such as: “Who's available?” “Call Anna.” “Send Peter a message.” “I'd like someone to talk to.” - And measure: How many interviewees rate this method at least 5 on a 1-7 scale for ease of use. - We are right if: At least 75% rate the voice-controlled interaction 5 or higher.

Test Card 6 Value Map	Assumption: The robot dog can create new social connections, not only maintain existing ones - We believe that: Some elderly people with limited mobility would be interested in meeting and talking with other elderly people who share similar interests, provided that the service feels safe and trustworthy. - To verify that, we will: Present a concept where the robot dog could say: “There are three people having a coffee chat about gardening. Would you like to join?” or: “Another person who enjoys fishing is available for a conversation. Would you like me to introduce you?” - And measure: How many interviewees would be willing to try a conversation with another elderly person based on a shared interest. - We are right if: At least 50% say they would be willing to try the service.
Test Card 7 Value Map	Assumption: Elderly people want to contribute to relationships, not only receive attention - We believe that: Social connection becomes more meaningful when the elderly person can contribute something to other people's lives rather than only receiving calls, assistance and entertainment. - To verify that, we will: Ask elderly interviewees how they would feel about using the system to share stories, knowledge, advice, recipes, memories or short voice messages with family members and friends. - And measure: How many express interest in actively contributing content or helping someone through the system. - We are right if: At least 60% identify at least one way they would like to contribute or share something with another person.

Interviews with potential future users showed that...

Interviews with potential future users showed that loneliness is not only caused by difficulties in using phones or other technology. Several interviewees said that they sometimes wanted to talk to someone but chose not to call because they assumed their children, relatives, or friends were busy and did not want to disturb them. The majority of the interviewees thought 5000kr for a Robo-Dog was reasonable.

The interviews also showed that small and spontaneous interactions can be important. Interviewees appreciated the idea of receiving short voice messages, photographs, greetings, or brief calls during the day rather than depending only on longer scheduled phone calls or visits.

A particularly positive response was given to the idea of knowing when family members or friends are available. Several interviewees said they would feel more comfortable initiating a call if they knew beforehand that the other person had time to talk. A simple function such as asking, “Who is available?” was therefore considered useful.

The interviewees generally preferred simple voice interaction over navigating menus, applications, and contact lists. Being able to say “Call my daughter,” “Who's available?” or “I'd like someone to talk to” was perceived as easier and more natural. They also appreciated the “memory support feature”, e.g. to be reminded of a close one’s birthday and an encouragement to call to congratulate them.

There was also some interest in being connected with people outside the immediate family, particularly other elderly people with similar interests. However, interviewees emphasized that such connections would need to feel safe and trustworthy. Some preferred only communicating with people they already knew.

Another important finding was that elderly people do not only want to receive attention and assistance. They also want to feel useful and involved in the lives of others. Sharing stories, giving advice, answering questions from grandchildren, or sending messages to family members were examples that received positive reactions.

The interviewees were generally positive to interacting with the robot dog itself and had experience of connecting to other robots such as lawn mowers or vacuum cleaners. One interviewee said the most important was that the robot was soft to touch so she could cuddle with it.

The interviews suggest that the most valuable role of the robot dog may not be to replace human companionship by becoming an artificial friend. Instead, its value could be in creating small moments of human connection throughout the day and making it easier for the elderly person to reach, interact with, and contribute to the lives of other people.

AI interview gave an overall introduction to different aspects to consider from users perspective:

Key Blindspots & Friction Points

A. Dignity, Stigma, and the "Infantilization" Barrier

- **User Perspective:** Being given a plush or robotic dog to replace human contact or a smartphone can feel condescending. Many older adults fear being treated like children or viewed as incompetent by family and care providers.
- **Missing Element:** The profile currently assumes people *want* to talk directly through a robot dog instead of a phone. From a user standpoint, speaking into a toy dog to make a phone call might feel bizarre or undignified unless the framing emphasizes empowerment and utility over a "toy" aesthetic.
- **Adjustment Needed:** Reframe the robot's role from a substitute for human interaction to an **enabler of independence and dignity.** Ensure the physical design and voice interactions feel mature, respectful, and high-tech rather than juvenile.

B. Voice-Only Control Realities

- **User Perspective:** The canvas assumes that eliminating buttons and relying 100% on voice control solves dexterity and vision issues. In practice, voice interaction brings its own heavy friction: dialect variations, speech changes due to age or stroke, memory loss regarding proper command phrases, and ambient background noise (e.g., TV).
- **Missing Element:** There is no backup input method or visual feedback mechanism mentioned.
- **Adjustment Needed:** Add **tactile/haptic feedback** (e.g., gentle vibrations, soft touch sensors) and subtle visual cues (like a gentle light collar) so users know the device is actually listening or working, without needing complex screens or buttons.

C. Privacy, Surveillance, and Autonomy

- **User Perspective:** Older adults who are cautious about digital tools are often hyper-aware of privacy. An always-listening device that tracks preferences, knows family contact lists, monitors location, and "nudges" them to go outside can feel invasive rather than supportive.
- **Missing Element:** Fear of loss of agency and surveillance is a major missing **Pain.**
- **Adjustment Needed:** Highlight clear **privacy guarantees** (e.g., local processing, no active eavesdropping) and **user control** (the user decides when the dog nudges them, rather than feeling managed by an AI).

D. Physical Weight, Maintenance, and Handling

- **User Perspective:** For users with reduced mobility or physical frailty, a robot dog that expects to be "taken outside" or "cuddled" introduces physical burdens.
- **Missing Element:** How is it charged? Is it heavy? What happens if it falls over or if the user trips over it?
- **Adjustment Needed:** Add **physical safety and low effort** to the Value Map. For instance, wireless/docking charging, light materials, and fall-prevention sensors.

Feedback on the Test Cards to consider:

- **Test Card 1 (Customer Profile - Family Willingness to Pay):** Testing family willingness to pay 5,000 SEK via Facebook ads is a solid quantitative metric. However, keep in mind that family members buying a product *for* an elderly relative does not guarantee the end-user will accept or adopt it.
- **Test Card 2 (Customer Profile - Talking through a dog):** Interviewing 10 elderly people on talking through a dog is crucial. Be prepared for resistance regarding the *form factor*; ask *why* they feel hesitant if they reject it (e.g., embarrassment vs. technical anxiety).
- **Test Card 2 (Value Map - Municipal Funding):** Expecting 40 out of 50 Swedish municipalities to reply to a survey is unrealistically high for public sector response rates. Aim for a sample of 15–20 in-depth interviews with municipal procurement/care managers instead to understand realistic budget cycles and procurement requirements.

Annex 4: Value Proposition Canvas (on the left – value map; on the right customer profile)